ICHOS: Spatial Audio & Contactless Gesture Station

AI-orchestrated music environment combining natural-language intent extraction, real-time live catalog search, synchronized karaoke lyrics, 60 FPS WebGL shaders, and 2-finger computer vision gesture control.

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1. EXECUTIVE OVERVIEW

ICHOS is a next-generation web-native spatial audio platform that bridges artificial intelligence with contactless human-computer interaction. Traditional music interfaces require constant manual input, rigid playlist structures, and physical device interaction. ICHOS eliminates every one of these friction points by pairing an **OpenAI-powered Intent Engine** (which translates moods, fictional characters, and cinematic scenes into acoustic search queries) with a **Zero-API-Key Direct YouTube Live Music Engine** that retrieves verified tracks in under 250 milliseconds. Audio is rendered through a **luxury minimal player** with synchronized karaoke lyrics, a **60 FPS Three.js WebGL cinema visualizer**, and a **Google MediaPipe computer vision pipeline** that allows users to switch tracks with a simple contactless 2-finger flick gesture.

2. THREE-TIER SYSTEM ARCHITECTURE

Ecosystem Tier	Core Technology Stack	Key Responsibilities
Frontend Client React 19 · Port 5173	Vite, React Router v7, Three.js (WebGL), Lucide Icons, Modern CSS Glassmorphism Modules	Luxury 2-column player with ambient glow artwork, headless persistent YouTube audio engine, 60 FPS WebGL shader cinema visualizer, real-time synchronized lyrics auto-scroller with click-to-seek, and contactless gesture HUD.
Backend Server Node.js · Port 5000	Express, OpenAI SDK (gpt-4o-mini structured outputs), LRCLIB Lyrics API, Native HTTPS Scraper	OpenAI Intent Engine translates prompts into 5-8 hyper-targeted YouTube queries. Live YouTube scraper with video-only filtering retrieves verified tracks without API keys. Candidate ranker/deduplicator assembles coherent 15-track queues. LRC timestamp parser powers karaoke.
Gesture Vision Python 3.10 · Port 8000	FastAPI, Uvicorn, OpenCV (cv2), Google MediaPipe Hands, WebSockets	Captures webcam at 30 FPS, extracts 21 3D skeletal hand landmarks, classifies 2-finger horizontal flick gestures via velocity thresholds (Δx > 0.05), and broadcasts real-time WebSocket events to the React frontend.

3. CORE TECHNICAL INNOVATIONS

■ OpenAI Intent Engine + Live YouTube Catalog

- Translates natural language prompts (e.g. *'Harvey Specter midnight blues'*, *'Barbie pink pop'*, *'Iron Man hard rock'*) into structured JSON mood profiles containing 5-8 hyper-targeted YouTube search strings.
- **Zero-Key Live Scraper** queries YouTube with `sp=EgIQAQ%253D%253D` (music video filter), cleaning title noise and extracting verified runtimes and format badges.
- Multi-variable ranker scores candidates against the intent vector and deduplicates into coherent 15-track queues.

■ Minimal Luxury Player + 60 FPS WebGL Cinema

- **Player Chamber:** 2-column luxury glass design — left card displays high-resolution album artwork with ambient blur glow; right card features a live 5-bar equalizer, bold title/artist, transport controls (shuffle, prev, play/pause, next, repeat), and volume slider.
- **Headless Audio Engine:** `autoplay: 0` prevents unwanted audio on page reload; audio streams persistently across all route transitions.
- **Cinema Mode:** 60 FPS Three.js WebGL procedural shader modulates radial light waves according to live bass, mid, and treble frequency energy.

■■ Synchronized Real-Time Karaoke Lyrics

- Sanitizes YouTube song titles and queries the open LRCLIB repository for timestamped `.lrc` lyrics, parsing them into structured `{time, text}` line arrays.
- **Karaoke Line Glow:** Active verse highlights in champagne gold and auto-centers smoothly in the viewport.
- **Click-to-Seek:** Clicking any lyric line instantly jumps audio playback to that exact timestamp.
- Floating lyric capsule rendered live in the Cinema Visualizer over the shader background.

■■ Contactless 2-Finger Flick Vision Engine

- Local Python FastAPI service processes webcam frames through Google MediaPipe, extracting 21 3D hand landmarks per frame at 30 FPS.
- **2-Finger Flick Classifier:** Detects Index + Middle finger extension and computes horizontal velocity — Δx > 0.05 triggers **Next Track**, Δx < –0.05 triggers **Previous Track**.
- Inverted flick correction and temporal debounce filters eliminate false positives.
- Events broadcast instantly over WebSocket to React PlayerContext for zero-latency track switching.

4. END-TO-END EXECUTION FLOWS

Pipeline	Step-by-Step Trace
Natural Language → Music Playback	① User enters a vibe prompt (e.g. *'Harvey Specter midnight drive'*). ② Frontend calls `POST /api/music/generate`. ③ OpenAI Intent Engine extracts mood, energy, genres, and formulates 5-8 targeted YouTube queries. ④ YouTube Catalog Engine executes queries in parallel on live YouTube. ⑤ Ranking Engine scores, deduplicates, and assembles a 15-track playlist. ⑥ Queue loads into PlayerContext → instant playback with synchronized lyrics.
Universal Real-Time Search	① User types any query in Search (e.g. *'Barbie'*, *'Taylor Swift'*). ② Search.jsx debounces (280ms) and calls `GET /api/music/search?q=…`. ③ youtubeService.js queries live YouTube with `sp=EgIQAQ%253D%253D`, returning clean titles, artist names, durations, and format badges. ④ User clicks Play or +Queue for instant playback.
Synchronized Karaoke Lyrics	① Track starts playing → Player.jsx requests `GET /api/music/lyrics`. ② lyricsService.js queries LRCLIB API and parses timestamped .lrc strings into line arrays. ③ As currentTime advances, active verse glows in champagne gold and auto-centers. ④ Clicking any line instantly seeks audio to that timestamp.
Contactless Camera Gesture	① User toggles Camera Gestures → WebSocket opens to `ws://localhost:8000/ws/gestures`. ② User performs a 2-finger horizontal flick in front of webcam. ③ MediaPipe extracts landmarks, computes Δx velocity, and emits `{"command":"NEXT"}`. ④ PlayerContext advances track with animated toast confirmation.
60 FPS WebGL Cinema Visualizer	① User navigates to Cinema (`/visualizer`). ② ReactiveVisualizer mounts Three.js ShaderAnimation. ③ WebGL fragment shader modulates radial wave amplitude and golden luminescence in real time based on bass, mid, and treble energy at a locked 60 FPS. ④ Active lyric lines float in an ambient glass capsule above the scrubber.

5. KEY FILE ARCHITECTURE

File	Layer	Purpose & Responsibility
`PlayerContext.jsx`	Client	Central state engine — manages active track, 15-song queue, play/pause, volume, shuffle, repeat, lyrics, and incoming gesture commands. Mounts headless YouTubePlayer at root level.
`Player.jsx`	Client	Luxury 2-column player chamber: album artwork with ambient glow, 5-bar equalizer, transport controls, dual-tab deck for Track Queue and Synchronized Lyrics with click-to-seek.
`Search.jsx`	Client	Real-time discovery: debounced live YouTube search, 9 trending vibe chips, recent search history, duration pills, format badges, and 1-click play/queue actions.
`shader-animation.jsx`	Client	60 FPS Three.js WebGL procedural shader with TWO_PI radial distortion loops reacting to bass (pulse), mid (ripples), and treble (glow).
`vibeService.js`	Server	OpenAI Intent Engine — analyzes prompts/images to extract mood, energy, genres, and generate 5-8 targetQueries. Includes 8 archetype heuristic fallbacks (Barbie, Iron Man, Harvey Specter, etc.).
`youtubeService.js`	Server	Live YouTube search engine — queries YouTube directly with video-only filters in <200ms without API keys. Cleans title noise, extracts durations, and applies format badges.
`lyricsService.js`	Server	LRCLIB Lyrics Engine — sanitizes titles, fetches synchronized .lrc timestamps, and parses them into {time, text} arrays for real-time karaoke rendering.
`app.py`	Vision	FastAPI + MediaPipe computer vision server — captures webcam at 30 FPS, extracts 21 hand landmarks, classifies 2-finger horizontal flick gestures, and streams events via WebSocket.

6. SYSTEM SPECIFICATIONS & DEPLOYMENT

Supported Inputs	Natural text prompts, character/mood/scene descriptions, image vision uploads, direct live keyword search, and trending vibe chip selection.
Audio Latency	< 250ms from search initiation to audio stream. WebGL cinema canvas locked at 60 FPS. Lyrics synchronization within ±100ms accuracy.
Gesture Latency	~33ms end-to-end (30 FPS MediaPipe pipeline → local WebSocket → React state dispatch). Temporal debounce window: 300ms.
1-Click Execution	Double-click `start-all.bat` in the project root to launch all 3 microservices and auto-open `http://localhost:5173` in the default browser.
Environment	Node.js 18+, Python 3.10+, any modern browser with WebGL support. Optional: `OPENAI_API_KEY` in server `.env` for AI-powered intent analysis (heuristic fallback available without key).